

SHEN LEMING

CONTACT INFORMATION

Affiliation: The Hong Kong Polytechnic University & UCL

E-mail: leming.shen@connect.polyu.hk

Phone: +852 62371304 +44 07756638932

Personal Homepage: <https://lemingshen.github.io/>

Research Interest: Robotics & Embodied AI, Large Language Models, Agents

EDUCATION

University College London, London, United Kingdom (U.S. News: 9)

2026.02 - 2026.07

Supervisor: Prof. Chris Xiaoxuan Lu

Visiting PhD in Robotics and Multimodal AI, Department of Computer Science

The Hong Kong Polytechnic University, Hong Kong, China (U.S. News: 52)

2022.08 - present

Supervisor: Prof. Yuanqing Zheng

PhD in Networking and Mobile Computing, Department of Computing

Zhejiang University, Hangzhou, China (U.S. News: 35)

2018.08 - 2022.06

Supervisor: Prof. Jinsong Han

Bachelor's Degree, Software Engineering, College of Computer Science and Technology

PUBLICATIONS

Conferences

- [SIGCOMM'26] Shiming Yu, **Leming Shen**, Jianing Zhang, Xin Li, Xianjin Xia, Yuanqing Zheng, and Yaxiong Xie. *RANPilot: Making AI Functionalities Robust to Dynamic O-RAN Reconfigurations*. In Proceedings of the ACM SIGCOMM 2026 Conference, 2026, Denver, Colorado, USA.
- [SenSys'26] Xinyu Huang, Qiang Yang, **Leming Shen**, Zijing Ma, and Yuanqing Zheng. *Jailbreaking Embodied LLMs via Action-level Manipulation*. In ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems, SenSys, May 11-14, 2026, Saint-Malo, France.
- [MobiCom'25] **Leming Shen**, Qiang Yang, Yuanqing Zheng, and Mo Li. *AutoIOT: LLM-Driven Automated Natural Language Programming for AIoT Applications*. In The 31st Annual International Conference on Mobile Computing and Networking, MobiCom, Nov. 4-8, 2025, Hong Kong, China.
Artifacts — Available  | Functional  | Reusable  | Replicated 
Project — <https://lemingshen.github.io/projects/autoiot/>
- [SenSys'25] **Leming Shen**, Qiang Yang, Xinyu Huang, Zijing Ma, and Yuanqing Zheng. *GPiOT: Tailoring Small Language Models for IoT Program Synthesis and Development*. In The 23rd ACM Conference on Embedded Networked Sensor Systems, SenSys, May 6-9, 2025, Irvine, USA.
Project — <https://lemingshen.github.io/projects/gpiot/>
- [MobiSys'24] **Leming Shen**, Qiang Yang, Kaiyan Cui, Yuanqing Zheng, Xiaoyong Wei, Jianwei Liu, and Jinsong Han. *FedConv: A Learning-on-Model Paradigm for Heterogeneous Federated Clients*. In Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services, MobiSys, June 3-7, Tokyo, Japan.
Artifacts — Available  | Functional  | Reusable 
Project — <https://lemingshen.github.io/projects/fedconv/>
- [INFOCOM'24] Kaiyan Cui, **Leming Shen**, Yuanqing Zheng, Fu Xiao, Jinsong Han. *Talk2Radar: Talking to mmWave Radars via Smartphone Speaker*. In Proceedings of the IEEE International Conference on Computer Communications, INFOCOM, May 20-23, 2024, Vancouver, Canada.

7. [SenSys'22] Ming Gao, Lingfeng Zhang, **Leming Shen**, Xiang Zou, Jinsong Han, Feng Lin, and Kui Ren. *KITE: Exploring the practical threat from acoustic transduction attacks on inertial sensors*. In Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems, SenSys, page 696–709, New York, NY, USA, 2023. Association for Computing Machinery.
8. [ICDCS'21] Jianwei Liu, Wenfan Song, **Leming Shen**, Jinsong Han, Xian Xu, Kui Ren, *MandiPass: Secure and Usable User Authentication via Earphone IMU*. 41st IEEE International Conference on Distributed Computing Systems, ICDCS, July 7-10, Washington DC, USA.

Journals

1. [arXiv] Xin Li, Shiming Yu, **Leming Shen**, Jianing Zhang, Yuanqing Zheng, Yaxiong Xie, *AutORAN: LLM-driven Natural Language Programming for Agile xApp Development*. arXiv preprint arXiv:2603.18604 (2026).
2. [TMC] **Leming Shen**, Qiang Yang, Kaiyan Cui, Yuanqing Zheng, Xiao-Yong Wei, Jianwei Liu, Jinsong Han, *Hierarchical and Heterogeneous Federated Learning via a Learning-on-Model Paradigm*. IEEE Transactions on Mobile Computing (2025).
3. [TMC] Xinyu Huang, **Leming Shen**, Zijing Ma, Yuanqing Zheng, *Towards Privacy-Preserving and Personalized Smart Homes via Tailored Small Language Models*. IEEE Transactions on Mobile Computing (2026).
4. [TMC] Kaiyan Cui, Qiang Yang, **Leming Shen**, Yuanqing Zheng, Fu Xiao, Jinsong Han, *Towards ISAC-Empowered mmWave Radars by Capturing Modulated Vibrations*. IEEE Transactions on Mobile Computing (2024).
5. [TMC] Ming Gao, Lingfeng Zhang, **Leming Shen**, Xiang Zou, Jinsong Han, Feng Lin, Kui Ren, *Exploring Practical Acoustic Transduction Attacks on Inertial Sensors in MDOF Systems*. IEEE Transactions on Mobile Computing (2023).
6. [TMC] Jianwei Liu, Wenfan Song, **Leming Shen**, Jinsong Han, Kui Ren, *Secure User Verification and Continuous Authentication via Earphone IMU*. IEEE Transactions on Mobile Computing (2023)

Short Papers

1. [MobiSys'26 Rising Star] **Leming Shen**, *Towards Automated, Resilient, and Robust AIoT*. In Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services, June 21–25, 2026, Cambridge, UK
2. [MobiSys'26 AutoSys Workshop] **Leming Shen**, Qiang Yang, Xinyu Huang, Zijing Ma, Yuanqing Zheng, and Chris Xiaoxuan Lu, *Automating Model Deployment on Mobile Devices*. In Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services, June 21–25, 2026, Cambridge, UK.
3. [MobiUK'26] **Leming Shen**, Qiang Yang, Xinyu Huang, Zijing Ma, Yuanqing Zheng, and Chris Xiaoxuan Lu, *Towards Automated Mobile Model Deployment*. In Eighth UK Mobile, Wearable and Ubiquitous Systems Research Symposium, June 25–26, 2026, Cambridge, UK.
4. [Getmobile'26] **Leming Shen**, Qiang Yang, Xinyu Huang, Zijing Ma, Yuanqing Zheng, and Chris Xiaoxuan Lu, *One Workflow Doesn't Fit All: Adaptive Workflows for Edge AI Development*. GetMobile: Mobile Computing and Communications 30, no. 1 (2026): 11-15.
5. [INFOCOM'26 EIN Workshop] **Leming Shen** and Yuanqing Zheng. *Federated Self-Evolving Embodied AI Agents*. In Proceedings of the IEEE International Conference on Computer Communications (INFOCOM) Workshop, Embodied Intelligence Networks (EIN), May 18–21, 2026, Tokyo, Japan.
6. [MobiCom'25] Xinyu Huang, **Leming Shen**, Zijing Ma, and Yuanqing Zheng. *Poster: Towards Privacy-Preserving and Personalized Smart Homes via Tailored Small Language Models*. In The 31st An-

nual International Conference on Mobile Computing and Networking (ACM MOBICOM '25), November 4–8, 2025, Hong Kong, China.

7. [CCS'25] Zijing Ma, **Leming Shen**, Xinyu Huang, and Yuanqing Zheng. *Poster: LLMalware: An LLM-Powered Robust and Efficient Android Malware Detection Framework*. In Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS '25), October 13–17, 2025, Taipei, Taiwan.
8. [MobiSys'25] **Leming Shen** and Yuanqing Zheng. *Poster: Towards Federated Embodied AI with FEAI*. In The 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom '24), September 30–October 4, 2024, Washington D.C., DC, USA.
9. [MobiCom'24] **Leming Shen**, and Yuanqing Zheng. *IoTCoder: A Copilot for IoT Application Development*. In Proceedings of the 30th Annual International Conference on Mobile Computing and Networking, MobiCom '24, Washington, D.C., USA.
10. [ICDCS'23] **Leming Shen**, Yuanqing Zheng, *PhD Symposium: FedDM: Data and Model Heterogeneity-Aware Federated Learning via Dynamic Weight Sharing*. 43rd IEEE International Conference on Distributed Computing Systems, ICDCS 2023, Hong Kong, China, July 18–21, 2023.
11. [SenSys'22] Kaiyan Cui, Qiang Yang, **Leming Shen**, Yuanqing Zheng, and Jinsong Han. *Poster Abstract: Integrated sensing and communication between daily devices and mmwave radars*. In Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems, SenSys '22, page 831–832, New York, NY, USA, 2023. Association for Computing Machinery.

PROFESSIONAL SERVICES

Organizing Committee

- Organizing committee of The 2nd Edition of Inscitech Meet on Robotics and Automation Technologies (IMROBOTECH2027)

TPC Member

- ACM UbiComp Workshop – EIoT 2026
- ACM UbiComp Demo, 2026
- The 21st International Conference on Green, Pervasive, and Cloud Computing (GPC 2026).
- IEEE MASS, 2026.
- ACM MobiSys Demos, 2026.
- ACM MobiSys Workshop – AutoSys, 2026.
- ACM MobiSys Workshop – HeadSys, 2026.
- ACM MobiCom Demos, 2025.
- ACM MobiCom Artifact Evaluation, 2025.
- ACM MobiSys Artifact Evaluation, 2024, 2025.
- ACM SenSys Artifact Evaluation, 2024.
- NDSS Artifact Evaluation, 2026.
- ACM CCS Artifact Evaluation, 2025.
- IEEE S&P Artifact Evaluation, 2026.
- USENIX Security Artifact Evaluation, 2026.

Invited Reviewer

- ACM Transactions on Internet of Things (TIOT) 2025.
- Computer Networks, 2025, 2026.
- IEEE Transactions on Mobile Computing (TMC) 2024.
- ACM Transactions on Sensor Networks (ToSN) 2024.
- IEEE Transactions on Wireless Communications (TWC) 2023, 2024.
- ACM IMWUT/UbiComp 2023, 2024, 2025.
- Expert Systems with Applications (ESWA), 2025, 2026
- Sensing and Bio-Sensing Research, 2025.
- Computers and Electrical Engineering, 2025, 2026.
- Computers and Electronics in Agriculture, 2026.
- Future Generation Computer Systems, 2026.
- SoftwareX, 2026.
- Neural Networks (NEUNET), 2026.
- Engineering Applications of Artificial Intelligence, 2026.
- Information & Communications Technology Express (ICT Express), 2026.

Others

- Microsoft Learn Student Ambassadors, 2022.
- Microsoft Student Partners, 2020, 2021.

HONORS & AWARDS

1. ACM MobiSys **Rising Star**, 2026
2. ACM CCS **Distinguished Artifact Evaluation Reviewer Award**, 2025.
3. ACM SenSys **Distinguished Artifact Evaluation Reviewer Award**, 2024.
4. ACM MobiSys **Distinguished Artifact Evaluation Reviewer Award**, 2024.
5. **Best Oral Presentation**, PolyU COMP - HKUST (GZ) INFH Research Student Conference, 2025.
6. ACM MobiSys Student Travel Grant, 2024
7. Outstanding Graduates of Zhejiang University, awarded on Undergraduate period, 2022.
8. Excellent Undergraduate Graduation Thesis of Zhejiang University, *Research and Implementation of Meta-learning-based Wi-Fi Activity Recognition System*, 2022

TALKS

- Towards Agentic AIoT Application Lifecycle @ King's College London, June 18, 2026.
- Towards Agentic AIoT Application Lifecycle @ Imperial College London, June 8, 2026.
- Towards Autonomous Development of Mobile Sensing Applications with Agentic LLMs @ The University of Cambridge, April 28, 2026.
- Towards Efficient Federated Learning on Edge @ Flower Lab Monthly, April 1, 2026.

TEACHING

- Teaching Assistant, COMP5434 Big Data Computing, 2024 Fall, 2025 Fall.
- Teaching Assistant, COMP5311 Internet Infrastructure and Protocols, 2022 Fall, 2023 Fall.
- Teaching Assistant, COMP1011 Programming Fundamentals, 2023 Spring, 2024 Spring, 2025 Spring.

SKILLS

- **Languages:** English, Mandarin
- **Programming:** Python, C++, C, C#, Java, MATLAB, Go, SQL, JavaScript, HTML, CSS, Pascal, LaTeX, Markdown, RST, JSON, XML
- **Platforms & Frameworks:** LangChain, Flower, Django, OpenAI, Vue, React, Flask

SOFTWARE PROJECT EXPERIENCE

- A LISP-like made-up programming language interpreter. (in Java)
- A MIPS assembly language interpreter. (in C#)
- A micro epidemic management web application. (in Python & Vue.js)
- A book injury system based on web spider. (in Java)
- A college teaching assistance web platform. (in Python & Bootstrap)
- An IoT device management and monitoring web platform. (in Python & React.js)
- A general merchandise vertical search engine. (in Java, Python & Vue.js)
- A distributed mini database system. (in C, C++, Java)